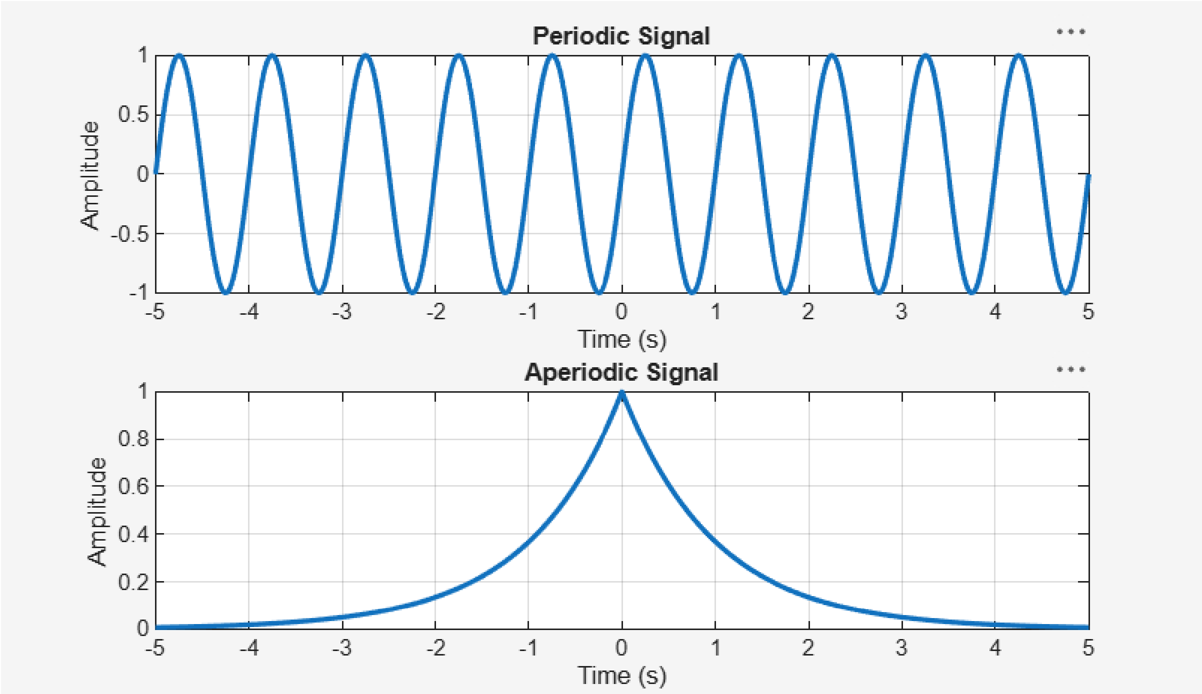
2 . SIGNAL PROPERTIES AND TRANSFORMATIONS

EXPT NO :2.1 Periodic and Aperiodic Signals

```
clc;
clear;
close all;
% Time axis
t = -5:0.01:5;
% Periodic Signal (Sine Wave)
x1 = sin(2*pi*t);
% Aperiodic Signal (Exponential Decay)
x2 = exp(-abs(t));
% Plotting
figure;
subplot(2,1,1);
plot(t,x1,'LineWidth',2);
grid on;
title('Periodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,1,2);
plot(t,x2,'LineWidth',2);
grid on;
title('Aperiodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

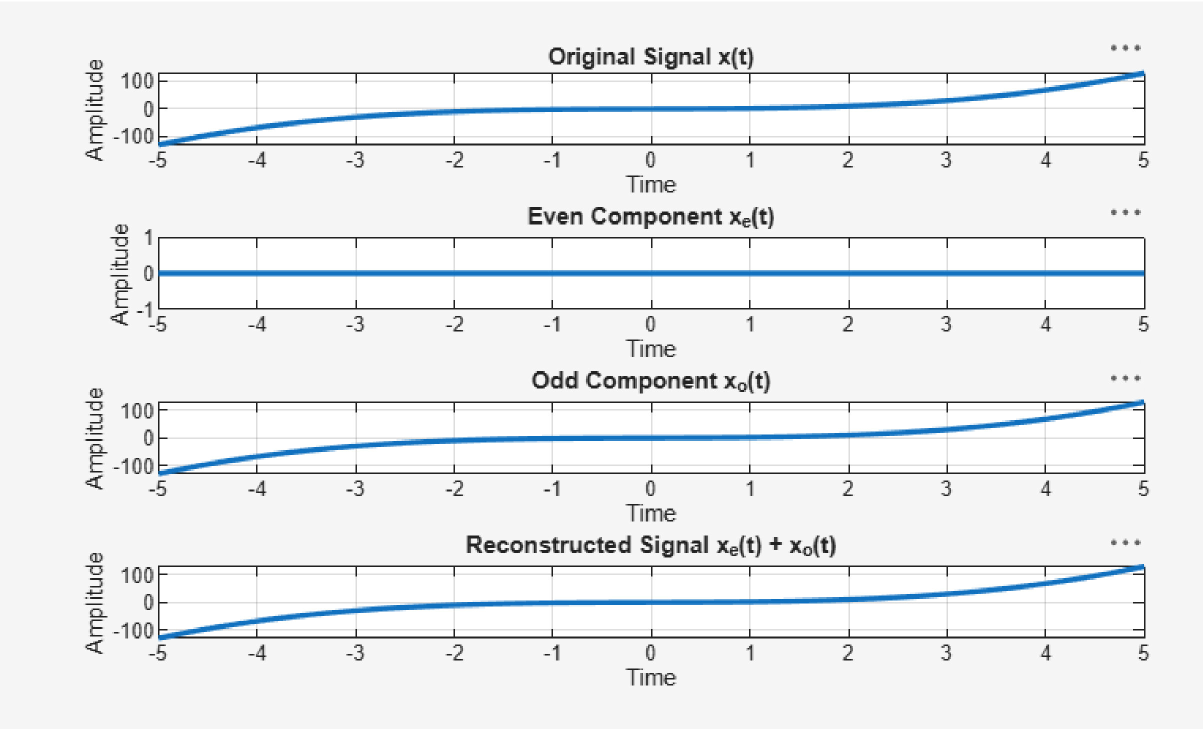

Periodic Signal
Amplitude
1
0.5
0
-0.5
-1
-5
-4
-3
-2
-1
0
1
2
3
4
5
Time (s)
Aperiodic Signal
Amplitude
1
0.8
0.6
0.4
0.2
0
-5
-4
-3
-2
-1
0
1
2
3
4
5
Time (s)

EXPT NO :2.2 Verification of Even and Odd Signal Decomposition

Program (MATLAB)

```
% Verification of Even and Odd Signal Decomposition
clc;
clear;
close all;
% Time axis
t = -5:0.1:5;
% Original Signal
x = t.^3 + t;
% Time-reversed Signal
x_rev = (-t).^3 + (-t);
% Even Component
xe = (x + x_rev)/2;
% Odd Component
xo = (x - x_rev)/2;
% Reconstruction of Original Signal
x_reconstructed = xe + xo;
% Plotting
figure;
subplot(4,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Original Signal x(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,2);
```

```
plot(t,xe,'LineWidth',2);
grid on;
title('Even Component x_e(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,3);
plot(t,xo,'LineWidth',2);
grid on;
title('Odd Component x_o(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,4);
plot(t,x_reconstructed,'LineWidth',2);
grid on;
title('Reconstructed Signal x_e(t) + x_o(t)');
xlabel('Time');
```




```
ylabel('Amplitude');
```